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fflags

Address: 0x0001

Size: 32 bits

Reset Value: 0x00000000

Description: Accrued Exceptions

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
nx	0	1	true	read-write	0x0	Inexact
uf	1	1	true	read-write	0x0	Underflow
of	2	1	true	read-write	0x0	Overflow
dz	3	1	true	read-write	0x0	Divide by Zero
nv	4	1	true	read-write	0x0	Invalid Operation
rsvd325	5	27	-	WPRI	-	Reserved

frm

Address: 0x0002

Size: 32 bits

Reset Value: 0x00000000

Description: Rounding Mode

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
frm	0	3	false	read-write	0x0	Rounding Mode
rsvd323	3	29	-	WPRI	-	Reserved

fcsr

Address: 0x0003

Size: 32 bits

Reset Value: 0x00000000

Description: FP control and status register

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
nx	0	1	true	read-write	0x0	Inexact
uf	1	1	true	read-write	0x0	Underflow
of	2	1	true	read-write	0x0	Overflow
dz	3	1	true	read-write	0x0	Divide by Zero
nv	4	1	true	read-write	0x0	Invalid Operation
frm	5	3	false	read-write	0x0	Rounding Mode
reserved	8	24	false	read-write	0x0	Reserved Bits

vstart

Address: 0x0008

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Vector start position

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
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vstart	0	8	true	WARL	0x0	Vector Start Index
rsvd648	8	56	-	WPRI	-	Reserved

vxsat

Address: 0x0009

Size: 32 bits

Reset Value: 0x00000000

Description: Fixed-point saturate flag

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
vxsat	0	1	true	WARL	0x0	Vector Fixed-Point Saturation Flag
rsvd321	1	31	-	WPRI	-	Reserved

vxrm

Address: 0x000A

Size: 32 bits

Reset Value: 0x00000000

Description: Fixed-point rounding mode

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
vxrm	0	2	true	WARL	0x0	Vector Fixed-Point Rounding Mode Register
rsvd322	2	30	-	WPRI	-	Reserved

vcsr

Address: 0x000F

Size: 32 bits

Reset Value: 0x00000000

Description: Vector control and status register

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
vxsat	0	1	true	WARL	0x0	Vector Fixed-Point Saturation Flag
vxrm	1	2	true	WARL	0x0	Vector Fixed-Point Rounding Mode Register
rsvd323	3	29	-	WPRI	-	Reserved

seed

Address: 0x0015

Size: 32 bits

Reset Value: 0x40000000

Description: Seed CSR.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
entropy	0	16	true	Read-Only	0x0	16b Entropy bits
custom	16	8	true	Read-Only	0x0	
rsvd_29_24	24	6	true	Read-Only	0x0	

opst	30	2	true	Read-Only	0x1	Register's Operating Status: 2'b01: WAIT, 2'b10: ES16, 2'b11: DEAD
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sstatus

Address: 0x0100

Size: 64 bits

Reset Value: 0x0000000200000000

Description: Supervisor status register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
sstatus_wpri_0	0	1	false	WPRI	0x0	WPRI
sie	1	1	false	WARL	0x0	interrupts are globally enabled when in S-mode
sstatus_wpri_1	2	3	false	WPRI	0x0	WPRI
spie	5	1	false	WARL	0x0	interrupt-enable bit active prior to the trap in S-mode
ube	6	1	false	WARL	0x0	Endianness: non-instruction-fetch memory accesses made from U-mode
sstatus_wpri_2	7	1	false	WPRI	0x0	WPRI
spp	8	1	false	WARL	0x0	previous privilege mode before entering S-mode
vs	9	2	false	WARL	0x0	status of the vector extension state
sstatus_wpri_3	11	2	false	WPRI	0x0	WPRI
fs	13	2	false	WARL	0x0	status of the floating-point unit state (Must become F-ext is implemented)
xs	15	2	false	WARL	0x0	status of additional user-mode extensions and associated state
sstatus_wpri_4	17	1	false	WPRI	0x0	WPRI
sum	18	1	false	WARL	0x0	permit Supervisor User Memory access
mrx	19	1	false	WARL	0x0	Make eXecutable Readable
sstatus_wpri_5	20	12	false	WPRI	0x0	WPRI
uxl	32	2	false	WARL	0x2	XLEN for U-mode
sstatus_wpri_6	34	29	false	WPRI	0x0	WPRI
sd	63	1	false	WARL	0x0	Summary of FS/VS/XS

sie

Address: 0x0104

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor interrupt-enable register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved

ssie	1	1	true	WARL	0x0	Interrupt-enable bit for supervisor-level Software Interrupts
hard0_0	2	2	true	WARL	0x0	Hardwired 0
rsvd44	4	1	-	WPRI	-	Reserved
stie	5	1	true	WARL	0x0	Interrupt-enable bit for supervisor-level Timer Interrupts
hard0_1	6	2	true	WARL	0x0	Hardwired 0
rsvd88	8	1	-	WPRI	-	Reserved
seie	9	1	true	WARL	0x0	Interrupt-enable bit for supervisor-level External Interrupts
hard0_2	10	3	true	WARL	0x0	Hardwired 0
lcofie	13	1	true	WARL	0x0	Interrupt-enable bit for supervisor-level Local Count Overflow Interrupts
rsvd6414	14	50	-	WPRI	-	Reserved

stvec

Address: 0x0105

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor trap handler base address.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mode_0	0	1	false	WARL	0x0	Allows mode 0 (base only) or mode 1 (vectored)
mode_1	1	1	false	WARL	0x0	Modes >= 2 are not allowed
basesxlen12 warl	2	62	false	WARL	0x0	Top SXLEN-2 bits of BASE

scounteren

Address: 0x0106

Size: 32 bits

Reset Value: 0x00000000

Description: Supervisor counter enable.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
cy	0	1	false	WARL	0x0	Cycle Counter Enable
tm	1	1	false	WARL	0x0	Time Counter Enable
ir	2	1	false	WARL	0x0	Instruction Retired Counter Enable
hpm3	3	1	false	WARL	0x0	Hardware Performance Monitoring Counter 3 Enable
hpm4	4	1	false	WARL	0x0	Hardware Performance Monitoring Counter 4 Enable
hpm5	5	1	false	WARL	0x0	Hardware Performance Monitoring Counter 5 Enable
hpm6	6	1	false	WARL	0x0	Hardware Performance Monitoring Counter 6 Enable
hpm7	7	1	false	WARL	0x0	Hardware Performance Monitoring Counter 7 Enable

hpm8	8	1	false	WARL	0x0	Hardware Performance Monitoring Counter 8 Enable
hpm9	9	1	false	WARL	0x0	Hardware Performance Monitoring Counter 9 Enable
hpm10	10	1	false	WARL	0x0	Hardware Performance Monitoring Counter 10 Enable
hpm11	11	1	false	WARL	0x0	Hardware Performance Monitoring Counter 11 Enable
hpm12	12	1	false	WARL	0x0	Hardware Performance Monitoring Counter 12 Enable
hpm13	13	1	false	WARL	0x0	Hardware Performance Monitoring Counter 13 Enable
hpm14	14	1	false	WARL	0x0	Hardware Performance Monitoring Counter 14 Enable
hpm15	15	1	false	WARL	0x0	Hardware Performance Monitoring Counter 15 Enable
hpm16	16	1	false	WARL	0x0	Hardware Performance Monitoring Counter 16 Enable
hpm17	17	1	false	WARL	0x0	Hardware Performance Monitoring Counter 17 Enable
hpm18	18	1	false	WARL	0x0	Hardware Performance Monitoring Counter 18 Enable
hpm19	19	1	false	WARL	0x0	Hardware Performance Monitoring Counter 19 Enable
hpm20	20	1	false	WARL	0x0	Hardware Performance Monitoring Counter 20 Enable
hpm21	21	1	false	WARL	0x0	Hardware Performance Monitoring Counter 21 Enable
hpm22	22	1	false	WARL	0x0	Hardware Performance Monitoring Counter 22 Enable
hpm23	23	1	false	WARL	0x0	Hardware Performance Monitoring Counter 23 Enable
hpm24	24	1	false	WARL	0x0	Hardware Performance Monitoring Counter 24 Enable
hpm25	25	1	false	WARL	0x0	Hardware Performance Monitoring Counter 25 Enable
hpm26	26	1	false	WARL	0x0	Hardware Performance Monitoring Counter 26 Enable
hpm27	27	1	false	WARL	0x0	Hardware Performance Monitoring Counter 27 Enable
hpm28	28	1	false	WARL	0x0	Hardware Performance Monitoring Counter 28 Enable
hpm29	29	1	false	WARL	0x0	Hardware Performance Monitoring Counter 29 Enable
hpm30	30	1	false	WARL	0x0	Hardware Performance Monitoring Counter 30 Enable
hpm31	31	1	false	WARL	0x0	Hardware Performance Monitoring Counter 31 Enable

senvcfg

Address: 0x010A

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor environment configuration register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
fiom	0	1	false	read-write	0x0	Fence of I/O implies Memory
wpri_0	1	3	false	WPRI	0x0	WPRI
cbie	4	2	false	read-write	0x0	Cache Block Invalidate instruction Enable — Enables the execution of CBO.INVALID in a lower privilege mode
cbcf	6	1	false	read-write	0x0	Cache Block Clean and Flush instruction enable — Enables execution of CBO.CLEAN and CBO.FLUSH in a lower privilege mode
cbze	7	1	false	read-write	0x0	Cache Block Zero instruction Enable — Enables execution of CBO.ZERO in a lower privilege mode
wpri_1	8	24	false	WPRI	0x0	WPRI
pmm	32	2	false	read-write	0x0	Enables pointer masking for the next lower privilege mode
wpri_2	34	30	false	WPRI	0x0	WPRI

sstateen0

Address: 0x010C

Size: 32 bits

Reset Value: 0x00000000

Description: Supervisor State Enable 0

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
c	0	1	false	WARL	0x0	Controls access to any and all custom state
fcsr	1	1	false	WARL	0x0	Controls access to fcsr CSR
jvt	2	1	false	WARL	0x0	Controls access to JVT CSR
wpri	3	29	false	WPRI	0x0	WPRI

sstateen1

Address: 0x010D

Size: 32 bits

Reset Value: 0x00000000

Description: Supervisor State Enable 1

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
wpri	0	32	false	WPRI	0x0	WPRI

sstateen2

Address: 0x010E

Size: 32 bits

Reset Value: 0x00000000

Description: Supervisor State Enable 2

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
wpri	0	32	false	WPRI	0x0	WPRI

sstateen3

Address: 0x010F

Size: 32 bits

Reset Value: 0x00000000

Description: Supervisor State Enable 3

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
wpri	0	32	false	WPRI	0x0	WPRI

sscratch

Address: 0x0140

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Scratch register for supervisor trap handlers.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
sscratch	0	64	false	read-write	0x0	Scratch register for supervisor trap handlers.

sepc

Address: 0x0141

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor exception program counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
addr	1	63	false	WARL	0x0	Supervisor exception program counter.

scause

Address: 0x0142

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor trap cause.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
exceptioncode ewlrl	0	63	false	read-write	0x0	Code identifying the last exception or interrupt
interrupt	63	1	false	read-write	0x0	Indicates if the trap was caused by an interrupt

stval

Address: 0x0143

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor Trap Value Register

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
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stval	0	64	false	WARL	0x0	Supervisor bad address or instruction.
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sip

Address: 0x0144

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor interrupt pending.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
ssip	1	1	true	WARL	0x0	Supervisor Software Interrupt Pending
hard0_1	2	2	true	WARL	0x0	Hardwired 0
rsvd44	4	1	-	WPRI	-	Reserved
stip	5	1	true	WARL	0x0	Supervisor Timer Interrupt Pending
hard0_2	6	2	true	WARL	0x0	Hardwired 0
rsvd88	8	1	-	WPRI	-	Reserved
seip	9	1	true	WARL	0x0	Supervisor External Interrupt Pending
hard0_3	10	3	true	WARL	0x0	Hardwired 0
lcofip	13	1	true	WARL	0x0	Supervisor Local Count Overflow Interrupt Pending
rsvd6414	14	50	-	WPRI	-	Reserved

stimecmp

Address: 0x014D

Size: 64 bits

Reset Value: 0x00000000FFFFFFFF

Description: Supervisor timer register

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
stimecmp	0	64	false	WARL	0xFFFFFFFF	Supervisor timer compare value

siselect

Address: 0x0150

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor Indirect register select (valid range is 0 - 0xFF)

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
interrupts	0	9	false	WARL	0x0	0x30 - 0x3F : Major Intr priorities 0x70-0xFF: External Intr. (0x71, 0x73-0x7F are rsvd) Rest: Reserved
rsvd_63_9	9	55	false	WARL	0x0	Reserved for future use

sireg

Address: 0x0151

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor indirect register alias

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
sireg	0	64	false	WARL	0x0	Supervisor indirect register alias

stopei

Address: 0x015C

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor top external interrupt (This is marked as read-write in AIA spec, but implemented as read-only for write to this CSR is ignored)

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
priority	0	11	true	WARL	0x0	Interrupt priority (same as identity)
rsvd_15_11	11	5	true	WARL	0x0	Reserved for future use
identity	16	11	true	WARL	0x0	Interrupt identity
rsvd_63_27	27	37	true	WARL	0x0	Reserved for future use

satp

Address: 0x0180

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor address translation and protection.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
ppn	0	44	false	WARL	0x0	Physical Page Number
asid	44	16	false	WARL	0x0	Address Space Identifier
mode	60	4	false	WARL	0x0	Address Translation and Protection Mode

srmcfg

Address: 0x0181

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor resource management qualification

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rcid	0	12	false	WARL	0x0	resource controller ID
rsvd1512	12	4	-	WPRI	-	Reserved
mcid	16	12	false	WARL	0x0	Monitor resource usage ID
rsvd6428	28	36	-	WPRI	-	Reserved

vsstatus

Address: 0x0200

Size: 64 bits

Reset Value: 0x0000000020000000

Description: Virtual supervisor status register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
vsstatus_wpr i_0	0	1	true	WPRI	0x0	WPRI
sie	1	1	true	WARL	0x0	interrupts are globally enabled when in VS-mode
vsstatus_wpr i_1	2	3	true	WPRI	0x0	WPRI
spie	5	1	true	WARL	0x0	interrupt-enable bit active prior to the trap in VS-mode
ube	6	1	true	WARL	0x0	Endianness: non-instruction-fetch memory accesses made from VU-mode
vsstatus_wpr i_2	7	1	true	WPRI	0x0	WPRI
spp	8	1	true	WARL	0x0	previous privilege mode before entering VS-mode
vs	9	2	true	WARL	0x0	status of the vector extension state
vsstatus_wpr i_3	11	2	true	WPRI	0x0	WPRI
fs	13	2	true	WARL	0x0	status of the floating-point unit state
xs	15	2	true	WARL	0x0	status of additional user-mode extensions and associated state
vsstatus_wpr i_4	17	1	true	WPRI	0x0	WPRI
sum	18	1	true	WARL	0x0	permit Supervisor User Memory access
mxr	19	1	true	WARL	0x0	Make eXecutable Readable
vsstatus_wpr i_5	20	12	true	WPRI	0x0	WPRI
uxl	32	2	true	WARL	0x2	XLEN for VU-mode
vsstatus_wpr i_6	34	29	true	WPRI	0x0	WPRI
sd	63	1	true	WARL	0x0	Summary of FS/VS/XS

vsie

Address: 0x0204

Size: 32 bits

Reset Value: 0x00000000

Description: Virtual supervisor interrupt-enable register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
vssie	1	1	false	read-write	0x0	Virtual Supervisor Software Interrupt Enable
hard0_0	2	2	false	read-write	0x0	Hardwired 0
rsvd44	4	1	-	WPRI	-	Reserved
vstie	5	1	false	read-write	0x0	Virtual Supervisor Timer Interrupt Enable
hard0_1	6	2	false	read-write	0x0	Hardwired 0
rsvd88	8	1	-	WPRI	-	Reserved
vseie	9	1	false	read-write	0x0	Virtual Supervisor External Interrupt Enable
hard0_2	10	3	false	read-write	0x0	Hardwired 0

lcofie	13	1	false	read-write	0x0	Virtual Supervisor Local Count Overflow Interrupt Enable
rsvd3214	14	18	-	WPRI	-	Reserved

vstvec

Address: 0x0205

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Virtual supervisor trap handler base address.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mode_0	0	1	false	WARL	0x0	Allows mode 0 (base only) or mode 1 (vectored)
mode_1	1	1	false	WARL	0x0	Modes >= 2 are not allowed
basesxlen12 warl	2	62	false	WARL	0x0	Top SXLEN-2 bits of BASE

vsscratch

Address: 0x0240

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Virtual supervisor scratch register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
sscratch	0	64	false	WARL	0x0	Virtual supervisor scratch register.

vsepc

Address: 0x0241

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Virtual supervisor exception program counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
addr	1	63	false	WARL	0x0	Virtual supervisor exception program counter.

vscause

Address: 0x0242

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Virtual supervisor trap cause.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
exceptioncode wlr	0	63	false	WARL	0x0	Code identifying the last exception or interrupt
interrupt	63	1	false	WARL	0x0	Indicates if the trap was caused by an interrupt

vstval

Address: 0x0243

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Virtualized supervisor bad guest physical address.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
vstval	0	64	false	WARL	0x0	Virtualized supervisor bad guest physical address.

vsip

Address: 0x0244

Size: 32 bits

Reset Value: 0x00000000

Description: Virtual supervisor interrupt pending.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
vssip	1	1	false	read-write	0x0	Virtual Supervisor Software Interrupt Pending
hard0_1	2	2	false	read-write	0x0	Hardwired 0
rsvd44	4	1	-	WPRI	-	Reserved
vstip	5	1	false	read-only	0x0	Virtual Supervisor Timer Interrupt Pending
hard0_2	6	2	false	read-write	0x0	Hardwired 0
rsvd88	8	1	-	WPRI	-	Reserved
vseip	9	1	false	read-only	0x0	Virtual Supervisor External Interrupt Pending
hard0_3	10	3	false	read-write	0x0	Hardwired 0
lcofip	13	1	false	read-only	0x0	Virtual Supervisor Local Count Overflow Interrupt Pending
rsvd3214	14	18	-	WPRI	-	Reserved

vstimecmp

Address: 0x024D

Size: 64 bits

Reset Value: 0x00000000FFFFFFFF

Description: Virtual supervisor timer register

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
vstimecmp	0	64	false	WARL	0xFFFFFFFF	Virtual supervisor timer compare value

vsiselect

Address: 0x0250

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Virtual supervisor Indirect register select (valid range is 0 - 0xFF)

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
interrupts	0	9	false	WARL	0x0	0x70-0xFF: External Intr. (0x71, 0x73-0x7F are rsvd)Rest: Inaccessible
rsvd_63_9	9	55	false	WARL	0x0	Reserved for future use

vsireg

Address: 0x0251

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Virtual supervisor indirect register alias

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
sireg	0	64	false	WARL	0x0	Virtual supervisor indirect register alias

vstopei

Address: 0x025C

Size: 64 bits

Reset Value: 0x0000000000000000

Description: This is Virtual supervisor top external interrupt (This is marked as read-write in AIA spec, but implemented as read-only for write to this CSR is ignored)

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
priority	0	11	true	WARL	0x0	Interrupt priority (same as identity)
rsvd_15_11	11	5	true	WARL	0x0	Reserved for future use
identity	16	11	true	WARL	0x0	Interrupt identity
rsvd_63_27	27	37	true	WARL	0x0	Reserved for future use

vsatp

Address: 0x0280

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Virtual Supervisor address translation and protection.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
ppn	0	44	false	WARL	0x0	Physical Page Number
asid	44	16	false	WARL	0x0	Address Space Identifier
mode	60	4	false	WARL	0x0	Address Translation and Protection Mode

mstatus

Address: 0x0300

Size: 64 bits

Reset Value: 0x0000000A00000000

Description: Machine status register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mstatus_wpr i_0	0	1	true	WPRI	0x0	WPRI

sie	1	1	true	WARL	0x0	hardware write: TRUE interrupts are globally enabled when in S-mode
mstatus_wpr i_1	2	1	true	WPRI	0x0	WPRI
mie	3	1	true	WARL	0x0	hardware write: TRUE interrupts are globally enabled when in M-mode
mstatus_wpr i_2	4	1	true	WPRI	0x0	WPRI
spie	5	1	true	WARL	0x0	hardware write: TRUE interrupt-enable bit active prior to the trap in S-mode
ube	6	1	true	WARL	0x0	Endianness: non-instruction-fetch memory accesses made from U-mode (not writable)
mpie	7	1	true	WARL	0x0	hardware write: TRUE interrupt-enable bit active prior to the trap in M-mode
spp	8	1	true	WARL	0x0	hardware write: TRUE previous privilege mode before entering S-mode
vs	9	2	true	WARL	0x0	status of the vector extension state
mpp	11	2	true	WARL	0x0	previous privilege mode before entering M-mode
fs	13	2	true	WARL	0x0	status of the floating-point unit state (Must become F-ext is implemented)
xs	15	2	true	WARL	0x0	status of additional user-mode extensions and associated state
mprv	17	1	true	WARL	0x0	Modify PRiVilege: modifies the effective privilege mode
sum	18	1	true	WARL	0x0	permit Supervisor User Memory access
mrx	19	1	true	WARL	0x0	Make eXecutable Readable
tvm	20	1	true	WARL	0x0	Trap Virtual Memory (should be writable for hypervisor emulation)
tw	21	1	true	WARL	0x0	Timeout Wait (should be writable for hypervisor emulation)
tsr	22	1	true	WARL	0x0	Trap SRET (FE catch illegal if SRET && TSR==1)
mstatus_wpr i_3	23	9	true	WPRI	0x0	WPRI
uxl	32	2	true	WARL	0x2	XLEN for U-mode
sxl	34	2	true	WARL	0x2	XLEN for S-mode
sbe	36	1	true	WARL	0x0	Endianness: non-instruction-fetch memory accesses made from S-mode
mbe	37	1	true	WARL	0x0	Endianness: non-instruction-fetch memory accesses made from M-mode
gva	38	1	true	WARL	0x0	Endianness: non-instruction-fetch memory accesses made from M-mode
mpv	39	1	true	WARL	0x0	Endianness: non-instruction-fetch memory accesses made from M-mode
mstatus_wpr i_4	40	23	true	WPRI	0x0	WPRI
sd	63	1	true	WARL	0x0	Summary of FS/VS/XS

misa

Address: 0x0301

Size: 64 bits

Reset Value: 0x8000000003411AF

Description: ISA and extensions

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
a	0	1	false	WARL	0x1	Atomic extension. Can be toggled to turn A extension on or off
b	1	1	true	Read-Only	0x1	Bit-manip Ext reflects cmisa.zb* fields. Can be toggled to turn B extension on or off
c	2	1	false	WARL	0x1	Compressed extension. Always supported
d	3	1	false	WARL	0x1	Double-precision FP extension. Can be toggled to turn D extension on or off
e	4	1	false	WARL	0x0	RV32E base ISA (Not allowing this)
f	5	1	false	WARL	0x1	Single-precision FP extension. Can be toggled to turn F extension on or off
g	6	1	false	WARL	0x0	Reserved
h	7	1	false	WARL	0x1	Hypervisor extension. Can be toggled to turn H extension on or off
i	8	1	false	WARL	0x1	RV64I base ISA. Always supported
j	9	1	false	WARL	0x0	Reserved
k	10	1	false	WARL	0x0	Reserved
l	11	1	false	WARL	0x0	Reserved
m	12	1	false	WARL	0x1	Integer Mul/Div extension. Can be toggled to turn M extension on or off
n	13	1	false	WARL	0x0	User-level interrupts extension (Do we need this?)
o	14	1	false	WARL	0x0	Reserved
p	15	1	false	WARL	0x0	Reserved
q	16	1	false	WARL	0x0	Quad precision FP extension (Not doing this)
r	17	1	false	WARL	0x0	Reserved
s	18	1	false	WARL	0x1	Supervisor mode implemented. Always supported
t	19	1	false	WARL	0x0	Reserved
u	20	1	false	WARL	0x1	User mode implemented. Always supported
v	21	1	false	WARL	0x1	Vector extension. Can be toggled to turn V extension on or off
w	22	1	false	WARL	0x0	Reserved
x	23	1	false	WARL	0x0	Non-standard extensions present (Saying No since, Not sure what this contains)
y	24	1	false	WARL	0x0	Reserved
z	25	1	false	WARL	0x0	Reserved
wlrl0	26	36	false	WARL	0x0	Hardwired 0
mxl	62	2	false	WARL	0x2	Native base integer ISA width

medeleg

Address: 0x0302

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine exception delegation register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
medeleg_0	0	10	false	WARL	0x0	Exception codes 0-9
medeleg_masked_0	10	1	false	WARL	0x0	Exception code 10
ecall_from_m	11	1	false	WARL	0x0	Exception code 11
medeleg_1	12	2	false	WARL	0x0	Exception codes 12-13
rsvd_0	14	1	false	WARL	0x0	Reserved exception code
medeleg_2	15	1	false	WARL	0x0	Exception code 15
rsvd_1	16	4	false	WARL	0x0	Reserved exception codes
medeleg_3	20	4	false	WARL	0x0	Exception codes 20-23
rsvd_2	24	40	false	WARL	0x0	Reserved exception codes

mideleg

Address: 0x0303

Size: 64 bits

Reset Value: 0x0000000000001444

Description: Machine interrupt delegation register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
ssip	1	1	false	WARL	0x0	Delegate Supervisor Software Interrupt
vssip	2	1	false	WARL	0x1	Delegate Virtual Supervisor Software Interrupt
msip	3	1	false	WARL	0x0	Delegate Machine Software Interrupt
rsvd44	4	1	-	WPRI	-	Reserved
stip	5	1	false	WARL	0x0	Delegate Supervisor Timer Interrupt
vstip	6	1	false	WARL	0x1	Delegate Virtual Supervisor Timer Interrupt
mtip	7	1	false	WARL	0x0	Delegate Machine Timer Interrupt
rsvd88	8	1	-	WPRI	-	Reserved
seip	9	1	false	WARL	0x0	Delegate Supervisor External Interrupt
vseip	10	1	false	WARL	0x1	Delegate Virtual Supervisor External Interrupt
meip	11	1	false	WARL	0x0	Delegate Machine External Interrupt
sgeip	12	1	false	WARL	0x1	Delegate Supervisor Guest External Interrupt
lcofip	13	1	false	WARL	0x0	Delegate Local Count Overflow Interrupt
rsvd6414	14	50	-	WPRI	-	Reserved

mie

Address: 0x0304

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine interrupt-enable register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
ssie	1	1	true	WARL	0x0	Interrupt-enable bit for Supervisor Software Interrupts
vssie	2	1	true	WARL	0x0	Interrupt-enable bit for Virtual Supervisor Software Interrupts
msie	3	1	true	WARL	0x0	Interrupt-enable bit for Machine Software Interrupts
rsvd44	4	1	-	WPRI	-	Reserved
stie	5	1	true	WARL	0x0	Interrupt-enable bit for Supervisor Timer Interrupts
vstie	6	1	true	WARL	0x0	Interrupt-enable bit for Virtual Supervisor Timer Interrupts
mtie	7	1	true	WARL	0x0	Interrupt-enable bit for Machine Timer Interrupts
rsvd88	8	1	-	WPRI	-	Reserved
seie	9	1	true	WARL	0x0	Interrupt-enable bit for Supervisor External Interrupts
vseie	10	1	true	WARL	0x0	Interrupt-enable bit for Virtual Supervisor External Interrupts
meie	11	1	true	WARL	0x0	Interrupt-enable bit for Machine External Interrupts
sgeie	12	1	true	WARL	0x0	Interrupt-enable bit for Supervisor Guest External Interrupts
lcofie	13	1	true	WARL	0x0	Interrupt-enable bit for Local Count Overflow Interrupts
rsvd1514	14	2	-	WPRI	-	Reserved
nonstandard interrupts	16	48	true	WARL	0x0	Definition of many of these interrupt numbers is either custom or non-ratified. So, these bits must be flexible

mtvec

Address: 0x0305

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine trap-handler base address.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mode_0	0	1	false	WARL	0x0	Allows mode 0 (base only) or mode 1 (vectored)
mode_1	1	1	false	WARL	0x0	Modes >= 2 are not allowed
base	2	62	false	WARL	0x0	Top MXLEN-2 bits of BASE

mcounteren

Address: 0x0306

Size: 32 bits

Reset Value: 0x00000000

Description: Machine counter enable.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
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cy	0	1	false	WARL	0x0	Cycle Counter Enable
tm	1	1	false	WARL	0x0	Time Counter Enable
ir	2	1	false	WARL	0x0	Instruction Retired Counter Enable
hpm3	3	1	false	WARL	0x0	Hardware Performance Monitoring Counter 3 Enable
hpm4	4	1	false	WARL	0x0	Hardware Performance Monitoring Counter 4 Enable
hpm5	5	1	false	WARL	0x0	Hardware Performance Monitoring Counter 5 Enable
hpm6	6	1	false	WARL	0x0	Hardware Performance Monitoring Counter 6 Enable
hpm7	7	1	false	WARL	0x0	Hardware Performance Monitoring Counter 7 Enable
hpm8	8	1	false	WARL	0x0	Hardware Performance Monitoring Counter 8 Enable
hpm9	9	1	false	WARL	0x0	Hardware Performance Monitoring Counter 9 Enable
hpm10	10	1	false	WARL	0x0	Hardware Performance Monitoring Counter 10 Enable
hpm11	11	1	false	WARL	0x0	Hardware Performance Monitoring Counter 11 Enable
hpm12	12	1	false	WARL	0x0	Hardware Performance Monitoring Counter 12 Enable
hpm13	13	1	false	WARL	0x0	Hardware Performance Monitoring Counter 13 Enable
hpm14	14	1	false	WARL	0x0	Hardware Performance Monitoring Counter 14 Enable
hpm15	15	1	false	WARL	0x0	Hardware Performance Monitoring Counter 15 Enable
hpm16	16	1	false	WARL	0x0	Hardware Performance Monitoring Counter 16 Enable
hpm17	17	1	false	WARL	0x0	Hardware Performance Monitoring Counter 17 Enable
hpm18	18	1	false	WARL	0x0	Hardware Performance Monitoring Counter 18 Enable
hpm19	19	1	false	WARL	0x0	Hardware Performance Monitoring Counter 19 Enable
hpm20	20	1	false	WARL	0x0	Hardware Performance Monitoring Counter 20 Enable
hpm21	21	1	false	WARL	0x0	Hardware Performance Monitoring Counter 21 Enable
hpm22	22	1	false	WARL	0x0	Hardware Performance Monitoring Counter 22 Enable
hpm23	23	1	false	WARL	0x0	Hardware Performance Monitoring Counter 23 Enable
hpm24	24	1	false	WARL	0x0	Hardware Performance Monitoring Counter 24 Enable
hpm25	25	1	false	WARL	0x0	Hardware Performance Monitoring Counter 25 Enable
hpm26	26	1	false	WARL	0x0	Hardware Performance Monitoring Counter 26 Enable
hpm27	27	1	false	WARL	0x0	Hardware Performance Monitoring Counter 27 Enable

hpm28	28	1	false	WARL	0x0	Hardware Performance Monitoring Counter 28 Enable
hpm29	29	1	false	WARL	0x0	Hardware Performance Monitoring Counter 29 Enable
hpm30	30	1	false	WARL	0x0	Hardware Performance Monitoring Counter 30 Enable
hpm31	31	1	false	WARL	0x0	Hardware Performance Monitoring Counter 31 Enable

mvien

Address: 0x0308

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine virtual interrupt enables

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hard0_0	0	1	false	WARL	0x0	Hardwired 0
ssip	1	1	false	WARL	0x0	Enable virtual interrupts for supervisor software interrupts
hard0_1	2	7	false	WARL	0x0	Hardwired 0
seip	9	1	false	WARL	0x0	Enable virtual interrupts for supervisor external interrupts
hard0_2	10	3	false	WARL	0x0	Hardwired 0
lcofip	13	1	false	WARL	0x0	Enable virtual interrupts for local count overflow interrupts
rsvd6414	14	50	-	WPRI	-	Reserved

mvip

Address: 0x0309

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine virtual interrupt pending bits

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hard0_0	0	1	true	WARL	0x0	Hardwired 0
ssip	1	1	true	WARL	0x0	Machine virtual supervisor software interrupt pending
hard0_1	2	3	true	WARL	0x0	Hardwired 0
stip	5	1	true	WARL	0x0	Machine virtual supervisor timer interrupt pending
hard0_2	6	3	true	WARL	0x0	Hardwired 0
seip	9	1	true	WARL	0x0	Machine virtual supervisor external interrupt pending
hard0_3	10	3	true	WARL	0x0	Hardwired 0
lcofip	13	1	true	WARL	0x0	Machine virtual local count overflow interrupt pending
rsvd6414	14	50	-	WPRI	-	Reserved

menvcfg

Address: 0x030A

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine environment configuration register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
fiom	0	1	false	WARL	0x0	Fence of I/O implies Memory
wpri_0	1	3	false	WPRI	0x0	WPRI
cbie	4	2	false	WARL	0x0	Cache Block Invalidate instruction Enable — Enables the execution of CBO.INVALID in a lower privilege mode
cbcf	6	1	false	WARL	0x0	Cache Block Clean and Flush instruction enable — Enables execution of CBO.CLEAN and CBO.FLUSH in a lower privilege mode
cbze	7	1	false	WARL	0x0	Cache Block Zero instruction Enable — Enables execution of CBO.ZERO in a lower privilege mode
wpri_1	8	24	false	WPRI	0x0	WPRI
pmm	32	2	false	WARL	0x0	Enables pointer masking for the next lower privilege mode
wpri_2	34	27	false	WPRI	0x0	
hade	61	1	false	WARL	0x0	Enables hardware updating of PTE A/D bits during S-mode address translation
pbmte	62	1	false	WARL	0x0	Enables Svpbmt extension for S-mode and G-stage address translation
stce	63	1	false	WARL	0x0	Enables STimecmp for S-mode

mstateen0

Address: 0x030C

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine State Enable 0

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
c	0	1	false	WARL	0x0	Controls access to any and all custom state
fcsr	1	1	false	WARL	0x0	Controls access to fcsr CSR
jvt	2	1	false	WARL	0x0	Controls access to JVT CSR
wpri_0	3	52	false	WPRI	0x0	WPRI
srncfg	55	1	false	WARL	0x0	Controls access to Srncfg CSR
p1p13	56	1	false	WARL	0x0	Controls access to hedeleg CSR
context	57	1	false	WARL	0x0	Controls access to scontext and hcontext CSRs
imsic	58	1	false	WARL	0x0	Controls access to IMSIC state
aia	59	1	false	WARL	0x0	Controls access to all state introduced by the Ssaia extension
csrind	60	1	false	WARL	0x0	Controls access to siselect, sireg*, vsiselect, vsireg* CSRs
wpri_1	61	1	false	WPRI	0x0	WPRI

envcfg	62	1	false	WARL	0x0	Controls access to the henvcfg, henvcfgh, and senvcfg CSRs
se0	63	1	false	WARL	0x0	Controls access to the hstateen0, hstateen0h, and sstateen0 CSRs

mstateen1

Address: 0x030D

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine State Enable 1

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
wpri	0	63	false	WPRI	0x0	WPRI
se1	63	1	false	WARL	0x0	Controls access to the hstateen1, hstateen1h, and sstateen1 CSRs

mstateen2

Address: 0x030E

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine State Enable 2

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
wpri	0	63	false	WPRI	0x0	WPRI
se2	63	1	false	WARL	0x0	Controls access to the hstateen2, hstateen2h, and sstateen2 CSRs

mstateen3

Address: 0x030F

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine State Enable 3

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
wpri	0	63	false	WPRI	0x0	WPRI
se3	63	1	false	WARL	0x0	Controls access to the hstateen1, hstateen1h, and sstateen1 CSRs

mcountinhibit

Address: 0x0320

Size: 32 bits

Reset Value: 0x00000000

Description: Machine counter-inhibit register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
cy	0	1	false	WARL	0x0	Cycle Counter Inhibit
hard0	1	1	false	WARL	0x0	Read-only zero
ir	2	1	false	WARL	0x0	Instruction Retired Counter Inhibit

hpm3	3	1	false	WARL	0x0	Hardware Performance Monitoring Counter 3 Inhibit
hpm4	4	1	false	WARL	0x0	Hardware Performance Monitoring Counter 4 Inhibit
hpm5	5	1	false	WARL	0x0	Hardware Performance Monitoring Counter 5 Inhibit
hpm6	6	1	false	WARL	0x0	Hardware Performance Monitoring Counter 6 Inhibit
hpm7	7	1	false	WARL	0x0	Hardware Performance Monitoring Counter 7 Inhibit
hpm8	8	1	false	WARL	0x0	Hardware Performance Monitoring Counter 8 Inhibit
hpm9	9	1	false	WARL	0x0	Hardware Performance Monitoring Counter 9 Inhibit
hpm10	10	1	false	WARL	0x0	Hardware Performance Monitoring Counter 10 Inhibit
hpm11	11	1	false	WARL	0x0	Hardware Performance Monitoring Counter 11 Inhibit
hpm12	12	1	false	WARL	0x0	Hardware Performance Monitoring Counter 12 Inhibit
hpm13	13	1	false	WARL	0x0	Hardware Performance Monitoring Counter 13 Inhibit
hpm14	14	1	false	WARL	0x0	Hardware Performance Monitoring Counter 14 Inhibit
hpm15	15	1	false	WARL	0x0	Hardware Performance Monitoring Counter 15 Inhibit
hpm16	16	1	false	WARL	0x0	Hardware Performance Monitoring Counter 16 Inhibit
hpm17	17	1	false	WARL	0x0	Hardware Performance Monitoring Counter 17 Inhibit
hpm18	18	1	false	WARL	0x0	Hardware Performance Monitoring Counter 18 Inhibit
hpm19	19	1	false	WARL	0x0	Hardware Performance Monitoring Counter 19 Inhibit
hpm20	20	1	false	WARL	0x0	Hardware Performance Monitoring Counter 20 Inhibit
hpm21	21	1	false	WARL	0x0	Hardware Performance Monitoring Counter 21 Inhibit
hpm22	22	1	false	WARL	0x0	Hardware Performance Monitoring Counter 22 Inhibit
hpm23	23	1	false	WARL	0x0	Hardware Performance Monitoring Counter 23 Inhibit
hpm24	24	1	false	WARL	0x0	Hardware Performance Monitoring Counter 24 Inhibit
hpm25	25	1	false	WARL	0x0	Hardware Performance Monitoring Counter 25 Inhibit
hpm26	26	1	false	WARL	0x0	Hardware Performance Monitoring Counter 26 Inhibit
hpm27	27	1	false	WARL	0x0	Hardware Performance Monitoring Counter 27 Inhibit
hpm28	28	1	false	WARL	0x0	Hardware Performance Monitoring Counter 28 Inhibit
hpm29	29	1	false	WARL	0x0	Hardware Performance Monitoring Counter 29 Inhibit

hpm30	30	1	false	WARL	0x0	Hardware Performance Monitoring Counter 30 Inhibit
hpm31	31	1	false	WARL	0x0	Hardware Performance Monitoring Counter 31 Inhibit

mhpmevent3

Address: 0x0323

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent3	0	56	false	WARL	0x0	Event selector
reserved	56	2	false	WARL	0x0	Reserved for future use
vuinh	58	1	false	WARL	0x0	VU-mode counter inhibit
vsinh	59	1	false	WARL	0x0	VS-mode counter inhibit
uinh	60	1	false	WARL	0x0	U-mode counter inhibit
sinh	61	1	false	WARL	0x0	S/HS-mode counter inhibit
minh	62	1	false	WARL	0x0	M-mode counter inhibit
of	63	1	false	WARL	0x0	Overflow status and interrupt disable bit

mhpmevent4

Address: 0x0324

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent4	0	56	false	WARL	0x0	Event selector
reserved	56	2	false	WARL	0x0	Reserved for future use
vuinh	58	1	false	WARL	0x0	VU-mode counter inhibit
vsinh	59	1	false	WARL	0x0	VS-mode counter inhibit
uinh	60	1	false	WARL	0x0	U-mode counter inhibit
sinh	61	1	false	WARL	0x0	S/HS-mode counter inhibit
minh	62	1	false	WARL	0x0	M-mode counter inhibit
of	63	1	false	WARL	0x0	Overflow status and interrupt disable bit

mhpmevent5

Address: 0x0325

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent5	0	56	false	WARL	0x0	Event selector
reserved	56	2	false	WARL	0x0	Reserved for future use

vuinh	58	1	false	WARL	0x0	VU-mode counter inhibit
vsinh	59	1	false	WARL	0x0	VS-mode counter inhibit
uinh	60	1	false	WARL	0x0	U-mode counter inhibit
sinh	61	1	false	WARL	0x0	S/HS-mode counter inhibit
minh	62	1	false	WARL	0x0	M-mode counter inhibit
of	63	1	false	WARL	0x0	Overflow status and interrupt disable bit

mhpmevent6

Address: 0x0326

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent6	0	56	false	WARL	0x0	Event selector
reserved	56	2	false	WARL	0x0	Reserved for future use
vuinh	58	1	false	WARL	0x0	VU-mode counter inhibit
vsinh	59	1	false	WARL	0x0	VS-mode counter inhibit
uinh	60	1	false	WARL	0x0	U-mode counter inhibit
sinh	61	1	false	WARL	0x0	S/HS-mode counter inhibit
minh	62	1	false	WARL	0x0	M-mode counter inhibit
of	63	1	false	WARL	0x0	Overflow status and interrupt disable bit

mhpmevent7

Address: 0x0327

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent7	0	56	false	WARL	0x0	Event selector
reserved	56	2	false	WARL	0x0	Reserved for future use
vuinh	58	1	false	WARL	0x0	VU-mode counter inhibit
vsinh	59	1	false	WARL	0x0	VS-mode counter inhibit
uinh	60	1	false	WARL	0x0	U-mode counter inhibit
sinh	61	1	false	WARL	0x0	S/HS-mode counter inhibit
minh	62	1	false	WARL	0x0	M-mode counter inhibit
of	63	1	false	WARL	0x0	Overflow status and interrupt disable bit

mhpmevent8

Address: 0x0328

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
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mhpmevent8	0	56	false	WARL	0x0	Event selector
reserved	56	2	false	WARL	0x0	Reserved for future use
vuinh	58	1	false	WARL	0x0	VU-mode counter inhibit
vsinh	59	1	false	WARL	0x0	VS-mode counter inhibit
uinh	60	1	false	WARL	0x0	U-mode counter inhibit
sinh	61	1	false	WARL	0x0	S/HS-mode counter inhibit
minh	62	1	false	WARL	0x0	M-mode counter inhibit
of	63	1	false	WARL	0x0	Overflow status and interrupt disable bit

mhpmevent9

Address: 0x0329

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent9	0	56	false	WARL	0x0	Event selector
reserved	56	2	false	WARL	0x0	Reserved for future use
vuinh	58	1	false	WARL	0x0	VU-mode counter inhibit
vsinh	59	1	false	WARL	0x0	VS-mode counter inhibit
uinh	60	1	false	WARL	0x0	U-mode counter inhibit
sinh	61	1	false	WARL	0x0	S/HS-mode counter inhibit
minh	62	1	false	WARL	0x0	M-mode counter inhibit
of	63	1	false	WARL	0x0	Overflow status and interrupt disable bit

mhpmevent10

Address: 0x032A

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent10	0	56	false	WARL	0x0	Event selector
reserved	56	2	false	WARL	0x0	Reserved for future use
vuinh	58	1	false	WARL	0x0	VU-mode counter inhibit
vsinh	59	1	false	WARL	0x0	VS-mode counter inhibit
uinh	60	1	false	WARL	0x0	U-mode counter inhibit
sinh	61	1	false	WARL	0x0	S/HS-mode counter inhibit
minh	62	1	false	WARL	0x0	M-mode counter inhibit
of	63	1	false	WARL	0x0	Overflow status and interrupt disable bit

mhpmevent11

Address: 0x032B

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent1 1	0	64	false	WARL	0x0	Event selector

mhpmevent12

Address: 0x032C

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent1 2	0	64	false	WARL	0x0	Event selector

mhpmevent13

Address: 0x032D

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent1 3	0	64	false	WARL	0x0	Event selector

mhpmevent14

Address: 0x032E

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent1 4	0	64	false	WARL	0x0	Event selector

mhpmevent15

Address: 0x032F

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent1 5	0	64	false	WARL	0x0	Event selector

mhpmevent16

Address: 0x0330

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent16	0	64	false	WARL	0x0	Event selector

mhpmevent17

Address: 0x0331

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent17	0	64	false	WARL	0x0	Event selector

mhpmevent18

Address: 0x0332

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent18	0	64	false	WARL	0x0	Event selector

mhpmevent19

Address: 0x0333

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent19	0	64	false	WARL	0x0	Event selector

mhpmevent20

Address: 0x0334

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent20	0	64	false	WARL	0x0	Event selector

mhpmevent21

Address: 0x0335

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent21	0	64	false	WARL	0x0	Event selector

mhpmevent22

Address: 0x0336

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent22	0	64	false	WARL	0x0	Event selector

mhpmevent23

Address: 0x0337

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent23	0	64	false	WARL	0x0	Event selector

mhpmevent24

Address: 0x0338

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent24	0	64	false	WARL	0x0	Event selector

mhpmevent25

Address: 0x0339

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent25	0	64	false	WARL	0x0	Event selector

mhpmevent26

Address: 0x033A

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent26	0	64	false	WARL	0x0	Event selector

mhpmevent27

Address: 0x033B

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent27	0	64	false	WARL	0x0	Event selector

mhpmevent28

Address: 0x033C

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent28	0	64	false	WARL	0x0	Event selector

mhpmevent29

Address: 0x033D

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent29	0	64	false	WARL	0x0	Event selector

mhpmevent30

Address: 0x033E

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent30	0	64	false	WARL	0x0	Event selector

mhpmevent31

Address: 0x033F

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring Event selector.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mhpmevent31	0	64	false	WARL	0x0	Event selector

mscratch

Address: 0x0340

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Scratch register for machine trap handlers.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mscratch	0	64	false	WARL	0x0	Machine Scratch Register

mepc

Address: 0x0341

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine exception program counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
addr	1	63	false	WARL	0x0	Machine exception program counter.

mcause

Address: 0x0342

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine trap cause.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
exceptioncode ewlrl	0	63	false	WARL	0x0	Code identifying the last exception or interrupt
interrupt	63	1	false	WARL	0x0	Indicates if the trap was caused by an interrupt

mtval

Address: 0x0343

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine bad address or instruction or machine trap value register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
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mtval	0	64	false	WARL	0x0	Machine Trap Value Register
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mip

Address: 0x0344

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine interrupt pending.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
ssip	1	1	true	WARL	0x0	Supervisor Software Interrupt Pending
vssip	2	1	true	WARL	0x0	Virtual Supervisor Software Interrupt Pending
msip	3	1	true	WARL	0x0	Machine Software Interrupt Pending
rsvd44	4	1	-	WPRI	-	Reserved
stip	5	1	true	WARL	0x0	Supervisor Timer Interrupt Pending
vstip	6	1	true	WARL	0x0	Virtual Supervisor Timer Interrupt Pending
mtip	7	1	true	WARL	0x0	Machine Timer Interrupt Pending
rsvd88	8	1	-	WPRI	-	Reserved
seip	9	1	true	WARL	0x0	Supervisor External Interrupt Pending
vseip	10	1	true	WARL	0x0	Virtual Supervisor External Interrupt Pending
meip	11	1	true	WARL	0x0	Machine External Interrupt Pending
sgeip	12	1	true	WARL	0x0	Supervisor Guest External Interrupt Pending
lcofip	13	1	true	WARL	0x0	Local Count Overflow Interrupt Pending
rsvd1514	14	2	-	WPRI	-	Reserved
nonstandard interrupts	16	48	true	WARL	0x0	Definition of many of these interrupt numbers is either custom or non-ratified. So, these bits must be flexible

mtinst

Address: 0x034A

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine mode trap instruction (transformed).

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mtinst	0	64	false	WARL	0x0	Machine Trap Instruction Register

mtval2

Address: 0x034B

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine mode bad guest physical address.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
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mtval2	0	64	false	WARL	0x0	Machine Second Trap Value Register
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miselect

Address: 0x0350

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine Indirect register select (valid range is 0 - 0xFF)

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
interrupts	0	8	false	WARL	0x0	0x30 - 0x3F : Major Intr priorities 0x70-0xFF: External Intr. (0x71, 0x73-0x7F are rsvd) Rest: Reserved
rsvd_63_8	8	56	false	WARL	0x0	Reserved for future use

mireg

Address: 0x0351

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine indirect register alias

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mireg	0	64	false	WARL	0x0	Machine indirect register alias

mtpei

Address: 0x035C

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine top external interrupt (This is marked as read-write in AIA spec, but implemented as read-only for write to this CSR is ignored)

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
priority	0	11	true	WARL	0x0	Interrupt priority (same as identity)
rsvd_15_11	11	5	true	WARL	0x0	Reserved for future use
identity	16	11	true	WARL	0x0	Interrupt identity
rsvd_63_27	27	37	true	WARL	0x0	Reserved for future use

pmpcfg0

Address: 0x03A0

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
pmp0cfg_rwx	0	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP0
pmp0cfg_mode	3	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP0

pmp0cfg_rsvd	5	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP0
pmp0cfg_locked	7	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP0
pmp1cfg_rwx	8	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP1
pmp1cfg_mode	11	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP1
pmp1cfg_rsvd	13	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP1
pmp1cfg_locked	15	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP1
pmp2cfg_rwx	16	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP2
pmp2cfg_mode	19	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP2
pmp2cfg_rsvd	21	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP2
pmp2cfg_locked	23	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP2
pmp3cfg_rwx	24	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP3
pmp3cfg_mode	27	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP3
pmp3cfg_rsvd	29	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP3
pmp3cfg_locked	31	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP3
pmp4cfg_rwx	32	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP4
pmp4cfg_mode	35	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP4
pmp4cfg_rsvd	37	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP4
pmp4cfg_locked	39	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP4
pmp5cfg_rwx	40	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP5
pmp5cfg_mode	43	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP5
pmp5cfg_rsvd	45	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP5
pmp5cfg_locked	47	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP5
pmp6cfg_rwx	48	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP6
pmp6cfg_mode	51	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP6
pmp6cfg_rsvd	53	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP6
pmp6cfg_locked	55	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP6
pmp7cfg_rwx	56	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP7

pmp7cfg_mode	59	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP7
pmp7cfg_rsvd	61	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP7
pmp7cfg_locked	63	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP7

pmpcfg2

Address: 0x03A2

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
pmp8cfg_rwx	0	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP8
pmp8cfg_mode	3	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP8
pmp8cfg_rsvd	5	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP8
pmp8cfg_locked	7	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP8
pmp9cfg_rwx	8	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP9
pmp9cfg_mode	11	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP9
pmp9cfg_rsvd	13	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP9
pmp9cfg_locked	15	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP9
pmp10cfg_rwx	16	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP10
pmp10cfg_mode	19	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP10
pmp10cfg_rsvd	21	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP10
pmp10cfg_locked	23	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP10
pmp11cfg_rwx	24	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP11
pmp11cfg_mode	27	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP11
pmp11cfg_rsvd	29	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP11
pmp11cfg_locked	31	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP11
pmp12cfg_rwx	32	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP12
pmp12cfg_mode	35	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP12
pmp12cfg_rsvd	37	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP12

pmp12cfg_locked	39	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP12
pmp13cfg_rwx	40	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP13
pmp13cfg_mode	43	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP13
pmp13cfg_rsvd	45	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP13
pmp13cfg_locked	47	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP13
pmp14cfg_rwx	48	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP14
pmp14cfg_mode	51	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP14
pmp14cfg_rsvd	53	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP14
pmp14cfg_locked	55	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP14
pmp15cfg_rwx	56	3	false	WARL	0x0	PMP Config Register RWX (Read/Write/Execute) bits for PMP15
pmp15cfg_mode	59	2	false	WARL	0x0	PMP Config Register A (Address Matching Mode) bits for PMP15
pmp15cfg_rsvd	61	2	false	WARL	0x0	PMP Config Register Rsvd bits for PMP15
pmp15cfg_locked	63	1	false	WARL	0x0	PMP Config Register L (Locked) bit for PMP15

pmpaddr0

Address: 0x03B0

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP0 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP0 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr1

Address: 0x03B1

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP1 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP1 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr2

Address: 0x03B2

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP2 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP2 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr3

Address: 0x03B3

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP3 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP3 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr4

Address: 0x03B4

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP4 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP4 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr5

Address: 0x03B5

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP5 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP5 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr6

Address: 0x03B6

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP6 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP6 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr7

Address: 0x03B7

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP7 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP7 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr8

Address: 0x03B8

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP8 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP8 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr9

Address: 0x03B9

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP9 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP9 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr10

Address: 0x03BA

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP10 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP10 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr11

Address: 0x03BB

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP11 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP11 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr12

Address: 0x03BC

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP12 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP12 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr13

Address: 0x03BD

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP13 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP13 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr14

Address: 0x03BE

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP14 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP14 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

pmpaddr15

Address: 0x03BF

Size: 64 bits

Reset Value: 0x00000000000001FF

Description: Physical memory protection configuration.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
address_lo	0	9	false	WARL	0x1FF	PMP15 address bits 10:2
address_hi	9	45	false	WARL	0x0	PMP15 address bits 55:11
rsvd6454	54	10	-	WPRI	-	Reserved

hstatus

Address: 0x0600

Size: 64 bits

Reset Value: 0x0000000200000000

Description: Hypervisor status register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
wpri_0	0	5	true	WPRI	0x0	WPRI
vsbe	5	1	true	WARL	0x0	Controls Endianness of explicit memory accesses made from VS-mode
gva	6	1	true	WARL	0x0	Indicates a Guest Virtual Address is written into stval
spv	7	1	true	WARL	0x0	Supervisor Previous Virtualization Mode
spvp	8	1	true	WARL	0x0	Supervisor Previous Virtual Privilege
hu	9	1	true	WARL	0x0	Controls whether HLV, HLVX, and HSV can be used in U-mode
wpri_1	10	2	true	WPRI	0x0	WPRI
vgein	12	6	true	WARL	0x0	Selects a guest external interrupt source for VS-level external interrupts
wpri_2	18	2	true	WPRI	0x0	WPRI
vtvm	20	1	true	WARL	0x0	Makes SFENCE.VMA, SINVAL.VMA, and accesses to satp raise virtual instruction exceptions
vtw	21	1	true	WARL	0x0	Makes WFI in VS-mode raises virtual instruction exception if not completed within a time limit

vtsr	22	1	true	WARL	0x0	Makes SRET give virtual instruction exception in VS-mode
wpri_3	23	9	true	WPRI	0x0	WPRI
vsxl	32	2	true	WARL	0x2	Controls the effective XLEN for VS-mode
wpri_4	34	14	true	WPRI	0x0	WPRI
hupmm	48	2	true	WARL	0x0	Configures pointer masking according to ssnp extension
wpri_5	50	14	true	WPRI	0x0	WPRI

hedeleg

Address: 0x0602

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor exception delegation register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hedeleg_0	0	9	false	WARL	0x0	Exception Codes 0-9
hard0_0	9	3	false	WARL	0x0	Hardwired 0
hedeleg_1	12	2	false	WARL	0x0	Exception Codes 12-13
hard0_1	14	1	false	WARL	0x0	Hardwired 0
hedeleg_2	15	1	false	WARL	0x0	Exception Code 15
hedeleg_3	16	4	false	WARL	0x0	Exception Codes 16-19
hard0_2	20	4	false	WARL	0x0	Hardwired 0
hard0_3	24	40	false	WARL	0x0	Hardwired 0

hideleg

Address: 0x0603

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor interrupt delegation register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
hard0_0	1	1	false	WARL	0x0	Hardwired 0
vssip	2	1	false	WARL	0x0	Delegate Virtual Supervisor Software Interrupt
hard0_1	3	1	false	WARL	0x0	Hardwired 0
rsvd44	4	1	-	WPRI	-	Reserved
hard0_2	5	1	false	WARL	0x0	Hardwired 0
vstip	6	1	false	WARL	0x0	Delegate Virtual Supervisor Timer Interrupt
hard0_3	7	1	false	WARL	0x0	Hardwired 0
rsvd88	8	1	-	WPRI	-	Reserved
hard0_4	9	1	false	WARL	0x0	Hardwired 0
vseip	10	1	false	WARL	0x0	Delegate Virtual Supervisor External Interrupt
hard0_5	11	2	false	WARL	0x0	Hardwired 0

lcofip	13	1	false	WARL	0x0	Hardwired 0 — Delegate Local Count Overflow Interrupt
rsvd6414	14	50	-	WPRI	-	Reserved

hie

Address: 0x0604

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor interrupt-enable register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
hard0_0	1	1	false	WARL	0x0	Hardwired 0
vssie	2	1	false	WARL	0x0	Virtual Supervisor Software Interrupt Enable
hard0_1	3	1	false	WARL	0x0	Hardwired 0
rsvd44	4	1	-	WPRI	-	Reserved
hard0_2	5	1	false	WARL	0x0	Hardwired 0
vstie	6	1	false	WARL	0x0	Virtual Supervisor Timer Interrupt Enable
hard0_3	7	1	false	WARL	0x0	Hardwired 0
rsvd88	8	1	-	WPRI	-	Reserved
hard0_4	9	1	false	WARL	0x0	Hardwired 0
vseie	10	1	false	WARL	0x0	Virtual Supervisor External Interrupt Enable
hard0_5	11	1	false	WARL	0x0	Hardwired 0
sgeie	12	1	false	WARL	0x0	Supervisor Guest External Interrupt Enable
rsvd6413	13	51	-	WPRI	-	Reserved

htimedelta

Address: 0x0605

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Delta for VS/VU-mode timer.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
htimedelta	0	64	false	WARL	0x0	Delta value between time and VS/VU mode timer

hcounteren

Address: 0x0606

Size: 32 bits

Reset Value: 0x00000000

Description: Hypervisor counter enable.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
cy	0	1	false	WARL	0x0	Cycle Counter Enable
tm	1	1	false	WARL	0x0	Time Counter Enable

ir	2	1	false	WARL	0x0	Instruction Retired Counter Enable
hpm3	3	1	false	WARL	0x0	Hardware Performance Monitoring Counter 3 Enable
hpm4	4	1	false	WARL	0x0	Hardware Performance Monitoring Counter 4 Enable
hpm5	5	1	false	WARL	0x0	Hardware Performance Monitoring Counter 5 Enable
hpm6	6	1	false	WARL	0x0	Hardware Performance Monitoring Counter 6 Enable
hpm7	7	1	false	WARL	0x0	Hardware Performance Monitoring Counter 7 Enable
hpm8	8	1	false	WARL	0x0	Hardware Performance Monitoring Counter 8 Enable
hpm9	9	1	false	WARL	0x0	Hardware Performance Monitoring Counter 9 Enable
hpm10	10	1	false	WARL	0x0	Hardware Performance Monitoring Counter 10 Enable
hpm11	11	1	false	WARL	0x0	Hardware Performance Monitoring Counter 11 Enable
hpm12	12	1	false	WARL	0x0	Hardware Performance Monitoring Counter 12 Enable
hpm13	13	1	false	WARL	0x0	Hardware Performance Monitoring Counter 13 Enable
hpm14	14	1	false	WARL	0x0	Hardware Performance Monitoring Counter 14 Enable
hpm15	15	1	false	WARL	0x0	Hardware Performance Monitoring Counter 15 Enable
hpm16	16	1	false	WARL	0x0	Hardware Performance Monitoring Counter 16 Enable
hpm17	17	1	false	WARL	0x0	Hardware Performance Monitoring Counter 17 Enable
hpm18	18	1	false	WARL	0x0	Hardware Performance Monitoring Counter 18 Enable
hpm19	19	1	false	WARL	0x0	Hardware Performance Monitoring Counter 19 Enable
hpm20	20	1	false	WARL	0x0	Hardware Performance Monitoring Counter 20 Enable
hpm21	21	1	false	WARL	0x0	Hardware Performance Monitoring Counter 21 Enable
hpm22	22	1	false	WARL	0x0	Hardware Performance Monitoring Counter 22 Enable
hpm23	23	1	false	WARL	0x0	Hardware Performance Monitoring Counter 23 Enable
hpm24	24	1	false	WARL	0x0	Hardware Performance Monitoring Counter 24 Enable
hpm25	25	1	false	WARL	0x0	Hardware Performance Monitoring Counter 25 Enable
hpm26	26	1	false	WARL	0x0	Hardware Performance Monitoring Counter 26 Enable
hpm27	27	1	false	WARL	0x0	Hardware Performance Monitoring Counter 27 Enable
hpm28	28	1	false	WARL	0x0	Hardware Performance Monitoring Counter 28 Enable

hpm29	29	1	false	WARL	0x0	Hardware Performance Monitoring Counter 29 Enable
hpm30	30	1	false	WARL	0x0	Hardware Performance Monitoring Counter 30 Enable
hpm31	31	1	false	WARL	0x0	Hardware Performance Monitoring Counter 31 Enable

hgeie

Address: 0x0607

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor guest external interrupt-enable register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hard0_0	0	1	false	WARL	0x0	Read-only zeros
guestexterna linterruptsw arl	1	5	false	WARL	0x0	Ascalon supports up to 5 guest supervisor-level interrupt files
hard0_1	6	58	false	WARL	0x0	Read-only zeros

hvien

Address: 0x0608

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor virtual interrupt enables

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hard0_0	0	13	false	WARL	0x0	Hardwired 0
lcofip	13	1	false	WARL	0x0	Local Count Overflow Interrupt
rsvd6414	14	50	-	WPRI	-	Reserved

hvictl

Address: 0x0609

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor virtual interrupt control

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
iprio	0	8	false	WARL	0x0	Interrupt Priority
ipriom	8	1	false	WARL	0x0	IPRIO Mode
dpr	9	1	false	WARL	0x0	Default Priority Rank
rsvd1510	10	6	-	WPRI	-	Reserved
iid	16	6	false	WARL	0x0	Interrupt Identity
rsvd2922	22	8	-	WPRI	-	Reserved
vti	30	1	false	WARL	0x0	Virtual Trap Interrupt control
rsvd6431	31	33	-	WPRI	-	Reserved

henvcfg

Address: 0x060A

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor environment configuration register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
fiom	0	1	false	WARL	0x0	Fence of I/O implies Memory
wpri_0	1	3	false	WPRI	0x0	WPRI
cbie	4	2	false	WARL	0x0	Cache Block Invalidate instruction Enable — Enables the execution of CBO.INVALID in a lower privilege mode
cbcf	6	1	false	WARL	0x0	Cache Block Clean and Flush instruction enable — Enables execution of CBO.CLEAN and CBO.FLUSH in a lower privilege mode
cbze	7	1	false	WARL	0x0	Cache Block Zero instruction Enable — Enables execution of CBO.ZERO in a lower privilege mode
wpri_1	8	24	false	WPRI	0x0	WPRI
pmm	32	2	false	WARL	0x0	Enables pointer masking for VS-mode
wpri_2	34	27	false	WPRI	0x0	WPRI
hade	61	1	false	WARL	0x0	Enables hardware updating of PTE A/D bits during VS-stage address translation
pbmte	62	1	false	WARL	0x0	Enables Svpbmt extension for VS-stage address translation
vstce	63	1	false	WARL	0x0	Enables VSTimecmp for VS-mode

hstateen0

Address: 0x060C

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor State Enable 0

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
c	0	1	false	WARL	0x0	Controls access to any and all custom state
fcsr	1	1	false	WARL	0x0	Controls access to fcsr CSR
jvt	2	1	false	WARL	0x0	Controls access to JVT CSR
wpri_0	3	54	false	WPRI	0x0	WPRI
context	57	1	false	WARL	0x0	Controls access to the scontext CSR
imsic	58	1	false	WARL	0x0	Controls access to IMSIC state
aia	59	1	false	WARL	0x0	Controls access to all state introduced by the Ssaia extension
csrind	60	1	false	WARL	0x0	Controls access to siselect, sireg*, vsiselect, vsireg* CSRs
wpri_1	61	1	false	WPRI	0x0	WPRI
envcfg	62	1	false	WARL	0x0	Controls access to the senvcfg CSR
se0	63	1	false	WARL	0x0	Controls access to the sstateen0 CSR

hstateen1

Address: 0x060D

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor State Enable 1

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
wpri	0	63	false	WPRI	0x0	WPRI
se1	63	1	false	WARL	0x0	Controls access to the sstateen1 CSR

hstateen2

Address: 0x060E

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor State Enable 2

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
wpri	0	63	false	WPRI	0x0	WPRI
se2	63	1	false	WARL	0x0	Controls access to the sstateen2 CSR

hstateen3

Address: 0x060F

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor State Enable 3

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
wpri	0	63	false	WPRI	0x0	WPRI
se3	63	1	false	WARL	0x0	Controls access to the sstateen3 CSR

htval

Address: 0x0643

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor Trap Value Register

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
htval	0	64	false	WARL	0x0	Hypervisor Trap Value Register

hip

Address: 0x0644

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor interrupt pending.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved

hard0_0	1	1	false	WARL	0x0	Hardwired 0
vssip	2	1	false	WARL	0x0	Virtual Supervisor Software Interrupt Pending
hard0_1	3	1	false	WARL	0x0	Hardwired 0
rsvd44	4	1	-	WPRI	-	Reserved
hard0_2	5	1	false	WARL	0x0	Hardwired 0
vstip	6	1	false	WARL	0x0	Virtual Supervisor Timer Interrupt Pending
hard0_3	7	1	false	WARL	0x0	Hardwired 0
rsvd88	8	1	-	WPRI	-	Reserved
hard0_4	9	1	false	WARL	0x0	Hardwired 0
vseip	10	1	false	WARL	0x0	Virtual Supervisor External Interrupt Pending
hard0_5	11	1	false	WARL	0x0	Hardwired 0
sgeip	12	1	false	WARL	0x0	Supervisor Guest External Interrupt Pending
rsvd6413	13	51	-	WPRI	-	Reserved

hvip

Address: 0x0645

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor virtual interrupt pending.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
hard0_0	1	1	true	WARL	0x0	Hardwired 0
vssip	2	1	true	WARL	0x0	Virtual Supervisor Software Interrupt Pending
hard0_1	3	1	true	WARL	0x0	Hardwired 0
rsvd44	4	1	-	WPRI	-	Reserved
hard0_2	5	1	true	WARL	0x0	Hardwired 0
vstip_virt	6	1	true	WARL	0x0	Virtual Supervisor Timer Interrupt Pending
hard0_3	7	1	true	WARL	0x0	Hardwired 0
rsvd88	8	1	-	WPRI	-	Reserved
hard0_4	9	1	true	WARL	0x0	Hardwired 0
vseip_virt	10	1	true	WARL	0x0	Virtual Supervisor External Interrupt Pending
hard0_5	11	2	true	WARL	0x0	Hardwired 0
lcofip_virt	13	1	true	WARL	0x0	Local Count Overflow Interrupt Pending
rsvd6414	14	50	-	WPRI	-	Reserved

hviprio1

Address: 0x0646

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor VS-level interrupt priorities 1—Assumed to be 0 for Athena. Lives in MC for this limited implementation

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hviprio1	0	64	false	WARL	0x0	Hypervisor VS-level interrupt priorities 1

hviprio2

Address: 0x0647

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor VS-level interrupt priorities 2—Assumed to be 0 for Athena. Lives in MC for this limited implementation

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hviprio2	0	64	false	WARL	0x0	Hypervisor VS-level interrupt priorities 2

htinst

Address: 0x064A

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor trap instruction (transformed).

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
htinst	0	64	false	WARL	0x0	Hypervisor trap Instruction Register

hgatp

Address: 0x0680

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor guest address translation and protection.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
ppn	0	44	false	WARL	0x0	Physical Page Number
vmid	44	14	false	WARL	0x0	Virtual Machine Identifier
warl0	58	2	false	WARL	0x0	Hardwired 0
mode	60	4	false	WARL	0x0	Address Translation and Protection Mode

mnscratch

Address: 0x0740

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Resumable NMI scratch register

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mnscratch	0	64	false	WARL	0x0	Resumable NMI scratch register

mnepc

Address: 0x0741

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Resumable NMI program counter

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd00	0	1	-	WPRI	-	Reserved
addr	1	63	false	WARL	0x0	Resumable NMI program counter

mncause

Address: 0x0742

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Resumable NMI cause

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
exceptioncode	0	63	false	WARL	0x0	Code identifying the last interrupt
interrupt	63	1	false	WARL	0x0	Indicates if the trap was caused by an interrupt

mnstatus

Address: 0x0744

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Resumable NMI status

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd20	0	3	-	WPRI	-	Reserved
nmie	3	1	false	WARL	0x0	Non Maskable Interrupt Enable
rsvd64	4	3	-	WPRI	-	Reserved
mnpv	7	1	false	WARL	0x0	Previous Virtualization Mode
rsvd108	8	3	-	WPRI	-	Reserved
mnpp	11	2	false	WARL	0x0	Previous Privilege Mode
rsvd6413	13	51	-	WPRI	-	Reserved

mseccfg

Address: 0x0747

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine security configuration register.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mml	0	1	false	WARL	0x0	Machine Mode Lockdown
mmwp	1	1	false	WARL	0x0	Machine Mode Whitelist Policy
rlb	2	1	false	WARL	0x0	Rule Locking Bypass

wpri_0	3	5	false	WPRI	0x0	WPRI
useed	8	1	false	WARL	0x0	Controls access to seed CSR in U mode
sseed	9	1	false	WARL	0x0	Controls access to seed CSR in S or HS mode
wpri_1	10	22	false	WPRI	0x0	WPRI
pmm	32	2	false	WARL	0x0	Enables pointer masking for M-mode
wpri_2	34	30	false	WPRI	0x0	WPRI

tselect

Address: 0x07A0

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Trigger Select

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
index	0	64	false	WARL	0x0	These are Trigger Registers from Debug Spec, not complete

dcscr

Address: 0x07B0

Size: 64 bits

Reset Value: 0x0000000000000003

Description: Debug Control and Status

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
prv	0	2	false	WARL	0x3	Previous Privilege Mode
step	2	1	false	WARL	0x0	When set and not in Debug Mode, the hart will only execute a single instruction and then enter Debug Mode
nmip	3	1	true	WARL	0x0	When set, there is a Non-Maskable-Interrupt (NMI) pending for the hart
mprven	4	1	false	WARL	0x0	Enable mprv in mstatus taking effect
v	5	1	false	WARL	0x0	Previous Virtualization Mode
cause	6	3	false	WARL	0x0	Explains why Debug Mode was entered
stoptime	9	1	false	WARL	0x0	time CSR is frozen at the time that Debug Mode was entered
stopcount	10	1	false	WARL	0x0	Don't increment any hart-local counters while in Debug Mode or on ebreak instructions causing entry into Debug Mode
stepie	11	1	false	WARL	0x0	Enable Interrupts during single stepping with step set
ebreaku	12	1	false	WARL	0x0	ebreak instructions in U-mode enter Debug Mode
ebreaks	13	1	false	WARL	0x0	ebreak instructions in S-mode enter Debug Mode
hard0_1	14	1	false	WARL	0x0	Hardwired 0
ebreakm	15	1	false	WARL	0x0	ebreak instructions in M-mode enter Debug Mode

ebreakvu	16	1	false	WARL	0x0	ebreak instructions in M-mode enter Debug Mode
ebreakvs	17	1	false	WARL	0x0	ebreak instructions in M-mode enter Debug Mode
hard0_2	18	10	false	WARL	0x0	Hardwired 0
xdebugver	28	4	false	WARL	0x0	Version of Debug Specification implemented
rsvd6432	32	32	-	WPRI	-	Reserved

dpc

Address: 0x07B1

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Debug PC

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
dpc	0	64	false	WARL	0x0	Debug PC

DSCRATCH0

Address: 0x07B2

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Debug Scratch Register 0

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
dscratch0	0	64	false	WARL	0x0	Debug Scratch Register 0

DSCRATCH1

Address: 0x07B3

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Debug Scratch Register 1

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
dscratch1	0	64	false	WARL	0x0	Debug Scratch Register 1

c_matp

Address: 0x07C7

Size: 32 bits

Reset Value: 0x00000000

Description: Machine address translation and protection.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
swid	0	1	false	WARL	0x0	Secure World Identifier
rsvd321	1	31	-	WPRI	-	Reserved

mcycle

Address: 0x0B00

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine cycle counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
cycle	0	64	true	WARL	0x0	Counter value

minstret

Address: 0x0B02

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine instructions-retired counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
instret	0	64	true	WARL	0x0	Counter value

mhpmcounter3

Address: 0x0B03

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 3	0	64	true	WARL	0x0	Counter value

mhpmcounter4

Address: 0x0B04

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 4	0	64	true	WARL	0x0	Counter value

mhpmcounter5

Address: 0x0B05

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 5	0	64	true	WARL	0x0	Counter value

mhpmcounter6

Address: 0x0B06

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 6	0	64	true	WARL	0x0	Counter value

mhpmcounter7

Address: 0x0B07

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 7	0	64	true	WARL	0x0	Counter value

mhpmcounter8

Address: 0x0B08

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 8	0	64	true	WARL	0x0	Counter value

mhpmcounter9

Address: 0x0B09

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 9	0	64	true	WARL	0x0	Counter value

mhpmcounter10

Address: 0x0B0A

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 10	0	64	true	WARL	0x0	Counter value

mhpmcounter11

Address: 0x0B0B

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter11	0	64	false	WARL	0x0	Counter value

mhpmcounter12

Address: 0x0B0C

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter12	0	64	false	WARL	0x0	Counter value

mhpmcounter13

Address: 0x0B0D

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter13	0	64	false	WARL	0x0	Counter value

mhpmcounter14

Address: 0x0B0E

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter14	0	64	false	WARL	0x0	Counter value

mhpmcounter15

Address: 0x0B0F

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter15	0	64	false	WARL	0x0	Counter value

mhpmcounter16

Address: 0x0B10

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 16	0	64	false	WARL	0x0	Counter value

mhpmcounter17

Address: 0x0B11

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 17	0	64	false	WARL	0x0	Counter value

mhpmcounter18

Address: 0x0B12

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 18	0	64	false	WARL	0x0	Counter value

mhpmcounter19

Address: 0x0B13

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 19	0	64	false	WARL	0x0	Counter value

mhpmcounter20

Address: 0x0B14

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 20	0	64	false	WARL	0x0	Counter value

mhpmcounter21

Address: 0x0B15

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 21	0	64	false	WARL	0x0	Counter value

mhpmcounter22

Address: 0x0B16

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 22	0	64	false	WARL	0x0	Counter value

mhpmcounter23

Address: 0x0B17

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 23	0	64	false	WARL	0x0	Counter value

mhpmcounter24

Address: 0x0B18

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 24	0	64	false	WARL	0x0	Counter value

mhpmcounter25

Address: 0x0B19

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 25	0	64	false	WARL	0x0	Counter value

mhpmcounter26

Address: 0x0B1A

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 26	0	64	false	WARL	0x0	Counter value

mhpmcounter27

Address: 0x0B1B

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 27	0	64	false	WARL	0x0	Counter value

mhpmcounter28

Address: 0x0B1C

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 28	0	64	false	WARL	0x0	Counter value

mhpmcounter29

Address: 0x0B1D

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 29	0	64	false	WARL	0x0	Counter value

mhpmcounter30

Address: 0x0B1E

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 30	0	64	false	WARL	0x0	Counter value

mhpmcounter31

Address: 0x0B1F

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter31	0	64	false	WARL	0x0	Counter value

c_async_int_status

Address: 0x0BF2

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine Local Interrupt Status Register

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
pa	0	56	true	WARL	0x0	PA associated with the local interrupt
rsvd_0	56	7	true	WARL	0x0	Reserved
valid	63	1	true	WARL	0x0	Valid bit

cycle

Address: 0x0C00

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Cycle counter for RDCYCLE instruction.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
cycle	0	64	false	Read-Only	0x0	Counter value

time

Address: 0x0C01

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Timer for RDTIME instruction. Read-only shadow of mtime MMR.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
time	0	64	true	Read-Only	0x0	Counter value

instret

Address: 0x0C02

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Instructions-retired counter for RDINSTRET instruction.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
instret	0	64	false	Read-Only	0x0	Counter value

hpmcounter3

Address: 0x0C03

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter3	0	64	false	Read-Only	0x0	Counter value

hpmcounter4

Address: 0x0C04

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter4	0	64	false	Read-Only	0x0	Counter value

hpmcounter5

Address: 0x0C05

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter5	0	64	false	Read-Only	0x0	Counter value

hpmcounter6

Address: 0x0C06

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter6	0	64	false	Read-Only	0x0	Counter value

hpmcounter7

Address: 0x0C07

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter7	0	64	false	Read-Only	0x0	Counter value

hpmcounter8

Address: 0x0C08

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter8	0	64	false	Read-Only	0x0	Counter value

hpmcounter9

Address: 0x0C09

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter9	0	64	false	Read-Only	0x0	Counter value

hpmcounter10

Address: 0x0C0A

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter10	0	64	false	Read-Only	0x0	Counter value

hpmcounter11

Address: 0x0C0B

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter11	0	64	false	Read-Only	0x0	Counter value

hpmcounter12

Address: 0x0C0C

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter12	0	64	false	Read-Only	0x0	Counter value

hpmcounter13

Address: 0x0C0D

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter13	0	64	false	Read-Only	0x0	Counter value

hpmcounter14

Address: 0x0C0E

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter14	0	64	false	Read-Only	0x0	Counter value

hpmcounter15

Address: 0x0C0F

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter15	0	64	false	Read-Only	0x0	Counter value

hpmcounter16

Address: 0x0C10

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter16	0	64	false	Read-Only	0x0	Counter value

hpmcounter17

Address: 0x0C11

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter17	0	64	false	Read-Only	0x0	Counter value

hpmcounter18

Address: 0x0C12

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter18	0	64	false	Read-Only	0x0	Counter value

hpmcounter19

Address: 0x0C13

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter19	0	64	false	Read-Only	0x0	Counter value

hpmcounter20

Address: 0x0C14

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter20	0	64	false	Read-Only	0x0	Counter value

hpmcounter21

Address: 0x0C15

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter21	0	64	false	Read-Only	0x0	Counter value

hpmcounter22

Address: 0x0C16

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter22	0	64	false	Read-Only	0x0	Counter value

hpmcounter23

Address: 0x0C17

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 23	0	64	false	Read-Only	0x0	Counter value

hpmcounter24

Address: 0x0C18

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 24	0	64	false	Read-Only	0x0	Counter value

hpmcounter25

Address: 0x0C19

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 25	0	64	false	Read-Only	0x0	Counter value

hpmcounter26

Address: 0x0C1A

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 26	0	64	false	Read-Only	0x0	Counter value

hpmcounter27

Address: 0x0C1B

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 27	0	64	false	Read-Only	0x0	Counter value

hpmcounter28

Address: 0x0C1C

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 28	0	64	false	Read-Only	0x0	Counter value

hpmcounter29

Address: 0x0C1D

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 29	0	64	false	Read-Only	0x0	Counter value

hpmcounter30

Address: 0x0C1E

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 30	0	64	false	Read-Only	0x0	Counter value

hpmcounter31

Address: 0x0C1F

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Performance-monitoring counter.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hpmcounter 31	0	64	false	Read-Only	0x0	Counter value

vl

Address: 0x0C20

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Vector length

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
vl	0	64	false	Read-Only	0x0	Vector length

vtype

Address: 0x0C21

Size: 64 bits

Reset Value: 0x8000000000000000

Description: Vector data type register

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
vlmul	0	3	true	WARL	0x0	Vector register group multiplier (LMUL) setting
vsew	3	3	true	WARL	0x0	Selected element width (SEW) setting
vta	6	1	true	WARL	0x0	Vector tail agnostic
vma	7	1	true	WARL	0x0	Vector mask agnostic
reserved	8	55	true	WARL	0x0	Reserved if non-zero
vill	63	1	true	WARL	0x1	Illegal value if set

vlenb

Address: 0x0C22

Size: 64 bits

Reset Value: 0x0000000000000020

Description: VLEN/8 (vector register length in bytes)

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
vlenb	0	64	false	Read-Only	0x20	VLEN/8 (vector register length in bytes)

scountovf

Address: 0x0DA0

Size: 64 bits

Reset Value: 0x0000000000000000

Description: HPM Counter Overflow bits

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
scountovf	0	32	true	Read-Only	0x0	HPM Counter Overflow bits
rsvd6432	32	32	-	WPRI	-	Reserved

stopi

Address: 0x0DB0

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Supervisor top interrupt

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
iprio	0	8	true	Read-Only	0x0	Interrupt Priority
rsvd_15_8	8	8	true	Read-Only	0x0	Reserved
iid	16	12	true	Read-Only	0x0	Interrupt Identity
rsvd_63_28	28	36	true	Read-Only	0x0	Reserved

hgeip

Address: 0x0E12

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hypervisor guest external interrupt pending.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hard0	0	1	true	Read-Only	0x0	Read-only zero
guestexterna linterrupts	1	63	true	Read-Only	0x0	Hypervisor guest external interrupts pending bits

vstopi

Address: 0x0EB0

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Virtual supervisor top interrupt

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
iprio	0	8	true	Read-Only	0x0	Interrupt Priority
rsvd_15_8	8	8	true	Read-Only	0x0	Reserved
iid	16	12	true	Read-Only	0x0	Interrupt Identity
rsvd_63_28	28	36	true	Read-Only	0x0	Reserved

mvendorid

Address: 0x0F11

Size: 32 bits

Reset Value: 0x00000000

Description: Vendor ID.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
jedecparity	0	7	false	Read-Only	FIXME	Parity_ID: 0xA1 = 1010 0001 (MSB is an odd parity bit)
jedecbankm 1	7	4	false	Read-Only	FIXME	JEDEC Bank = 16. This represents Jedec Bank -1 . ([15][0x21 - 1] = Tenstorrent Inc.)
rsvd3211	11	21	-	WPRI	-	Reserved

marchid

Address: 0x0F12

Size: 32 bits

Reset Value: 0x00000000

Description: Architecture ID.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
sharedcache version	0	8	false	Read-Only	FIXME	Shared Cache Version
microarchve rsion	8	8	false	Read-Only	FIXME	Micro-Architecture Version
archversion	16	8	false	Read-Only	FIXME	Architecture Version

ipname	24	8	false	Read-Only	FIXME	IP Name
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mimpid

Address: 0x0F13

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Implementation ID.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
rsvd_31_0	0	12	false	Read-Only	0x0	Reserved
dropmaintversion	12	4	false	Read-Only	FIXME	Drop Maintenance Version
dropminorversion	16	6	false	Read-Only	FIXME	Drop Major Version
dropmajorversion	22	4	false	Read-Only	FIXME	Drop Major Version
droptype	26	2	false	Read-Only	FIXME	Drop Type
manid	28	4	false	Read-Only	FIXME	Man ID
stepmetalid	32	4	false	Read-Only	FIXME	Step Metal ID
stepbaseid	36	4	false	Read-Only	FIXME	Step Base ID
sharedcacheconfig	40	8	false	Read-Only	FIXME	Shared Cache Config
coreconfig	48	8	false	Read-Only	FIXME	Core Config
clusterconfig	56	8	false	Read-Only	FIXME	Cluster Config

mhartid

Address: 0x0F14

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Hardware thread ID. (Starts from 0 and increment for each core)

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
hartidincluster	0	3	false	Read-Only	0x0	Hart ID for the core in a cluster
clusterid	3	4	false	Read-Only	0x0	Cluster ID (defined by external pins)
rsvd_63_7	7	57	false	Read-Only	0x0	Reserved for future use

mconfigptr

Address: 0x0F15

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Pointer to configuration data structure.

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
mconfigptr	0	64	false	Read-Only	0x0	configuration pointer, may be hardwired

mtopi

Address: 0x0FB0

Size: 64 bits

Reset Value: 0x0000000000000000

Description: Machine top interrupt

Field Name	Bit Offset	Bit Width	Volatile	Access	Reset Value	Description
iprio	0	8	true	Read-Only	0x0	Interrupt Priority
rsvd_15_8	8	8	true	Read-Only	0x0	Reserved
iid	16	12	true	Read-Only	0x0	Interrupt Identity
rsvd_63_28	28	36	true	Read-Only	0x0	Reserved

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